Q6 (a).

$$I_1 = \int_{-3\pi/2}^{3\pi/2} e^{2x} x^2 \, dx.$$

We'll use integration by parts, maybe expecting to apply it twice.

$$I_1 = \int_{-3\pi/2}^{3\pi/2} \frac{d}{dx}\left(\frac{e^{2x}}{2}\right) x^2 \, dx$$, preparing the factors.

$$= \left[\frac{e^{2x}}{2} x^2 \right]_{-3\pi/2}^{3\pi/2} - \int_{-3\pi/2}^{3\pi/2} \frac{e^{2x}}{2} 2x \, dx.$$, int. by parts.

$$= \frac{1}{2}\left[e^{2x} x^2 \right]_{-3\pi/2}^{3\pi/2} - \int_{-3\pi/2}^{3\pi/2} \frac{d}{dx}\left(\frac{e^{2x}}{2}\right) x \, dx.$$

, preparing again for int. by parts.

$$= \frac{1}{2}\left[e^{2x} x^2 \right]_{-3\pi/2}^{3\pi/2} - \left[\frac{e^{2x}}{2} x \right]_{-3\pi/2}^{3\pi/2} + \int_{-3\pi/2}^{3\pi/2} \frac{e^{2x}}{2} \, dx.$$

, by int. by parts.

$$= \frac{1}{2}\left[e^{2x}x^2\right]_{-3\pi/2}^{3\pi/2} - \frac{1}{2}\left[e^{2x}\,x\right]_{-3\pi/2}^{3\pi/2}.$$

$$+\left[\frac{e^{2x}}{4}\right]_{-3\pi/2}^{3\pi/2}.$$

, using integral of the exponential function.

$$= e^{3\pi}\left(\frac{9\pi^2}{8} - \frac{3\pi}{4} + \frac{1}{4}\right)$$

$$- e^{-3\pi}\left(\frac{9\pi^2}{8} + \frac{3\pi}{4} + \frac{1}{4}\right)$$

$$= \frac{e^{3\pi}}{8}\left(9\pi^2 - 6\pi + 2\right)$$

$$- \frac{e^{-3\pi}}{8}\left(9\pi^2 + 6\pi + 2\right).$$

$$\int \frac{d}{dx}(f)\, g \; dx,$$

$$= \left[f(x)\, g(x) \right] - \int f(x) \frac{dg}{dx} \, dx$$

Prod rule.

$$\cancel{\int} \frac{d}{dx}\left[f(x)\, g(x) \right] = \int \frac{d}{dx}(f)\, g + \int f \frac{dg}{dx}.$$

Q5 (b)

$$g(x) = 2x^4 \cos(3x^2)$$

Applying the product rule first to get

$$g'(x) = 2x^4 \frac{d}{dx}\left(\cos(3x^2)\right) + \frac{d}{dx}(2x^4)\cos(3x^2)$$

, now using known derivatives of cosine and powers to get, and chain rule,

$$g'(x) = 2x^4\left(-\sin(3x^2)\right)6x + 8x^3\cos(3x^2).$$

$$= -12x^5 \sin(3x^2) + 8x^3\cos(3x^2).$$

Now for the second derivative we need

$$g''(x) = \frac{d}{dx}\left(g'(x)\right).$$

employing in order the linearity, product and ~~chain rule~~ mainly to get

$$g''(x) = -12x^5 \frac{d}{dx}\left(\sin(3x^2)\right) + \frac{d}{dx}(-12x^5)\sin(3x^2)$$

$$+ 8x^3 \frac{d}{dx}\left(\cos(3x^2)\right) + \frac{d}{dx}(8x^3)\cos(3x^2).$$

Next employing chain rule along with known derivatives of sine, cosine and powers

$$g''(x) = -12x^5 \cos(3x^2)\, 6x - 60x^4 \sin(3x^2)$$

$$+ 8x^3\left(-\sin(3x^2)\right) 6x + 24x^2\cos(3x^2)$$

$$= \cos(3x^2)\left(-72x^6 + 24x^2\right)$$

$$+ \sin(3x^2)\left(-108x^4\right)$$

(d) First applying the Quotient rule.

$$F'(x) = \frac{v(x)w(x)z(x)\,u'(x) - u(x)\frac{d}{dx}(vwz)}{\left(v(x)\,w(x)\,z(x)\right)^2}.$$

and now the, three-factor product rule, applied to the last factor in numerator

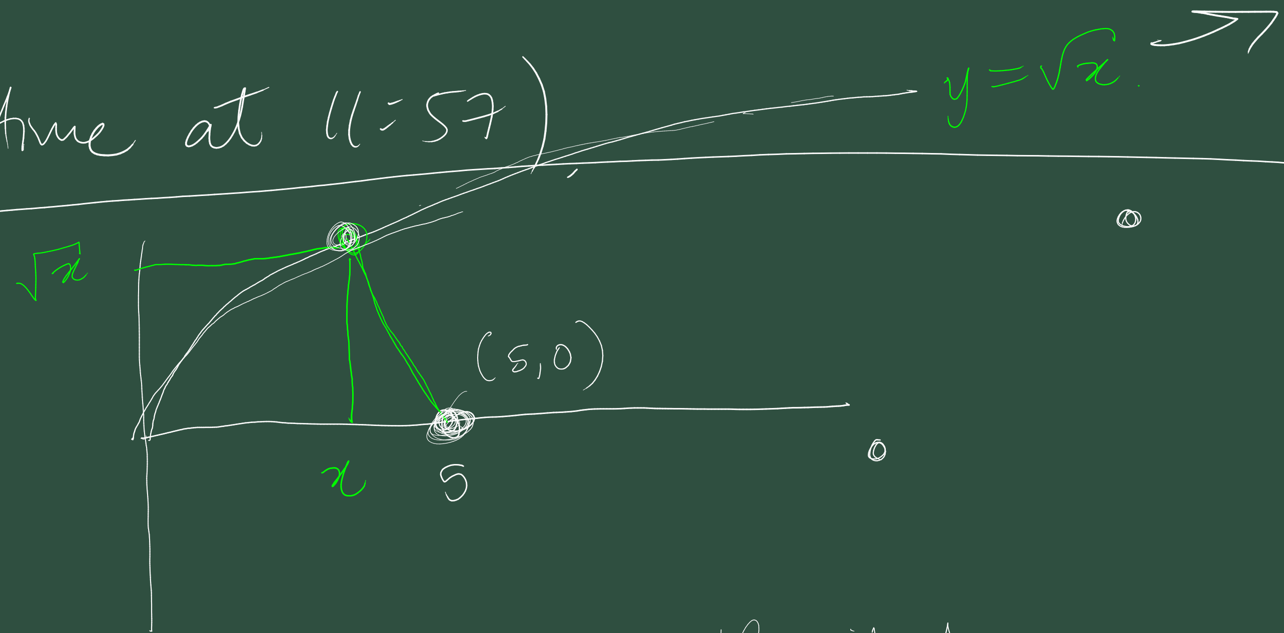
$$= \frac{u'(x)v(x)w(x)z(x) - u(x)\left(v'(x)w(x)z(x) + v(x)w'(x)z(x) + v(x)w(x)z'(x)\right)}{v^2(x)\,w^2(x)\,z^2(x)}$$

, suppressing most of the (x)s.

$$= \frac{u'vwz - uw'vz - uwv'z - uwvz'}{v^2(x)\,w^2(x)\,z^2(x)}$$

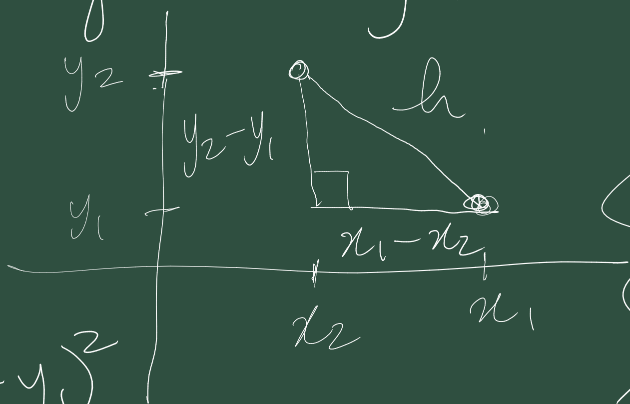
(continue at 11:57)

(c).



We need to minimise the distance function $D(x)$ which is the distance from the point $(x, \sqrt{x})$ to $(5,0)$. D is defined by.

$$\left\{ \begin{aligned} h^2 &= (x_1 - x_2)^2 + (y_2 - y_1)^2 \\ &= (x_2 - x_1)^2 + (y_2 - y_1)^2 \end{aligned} \right.$$



$$D(x) = \left((x-5)^2 + (\sqrt{x})^2 \right)^{1/2}$$

$$= \left(x^2 - 10x + 25 + x \right)^{1/2}$$

$$= \left(x^2 - 9x + 25 \right)^{1/2} \qquad (3)$$

~~Find stationary~~ points by ~~solu~~

Stationary points are the solutions to $D'(x) = 0$. (2)

$$D'(x) = \frac{1}{2}\left(x^2 - 9x + 25 \right)^{-1/2} (2x-9)$$

from using the power rule and chain rule for differentiating $\sqrt{u}$

$$D'(x) = \frac{2x-9}{2\sqrt{x^2-9x+25}} = 0.$$

This is satisfied when

$$2x - 9 = 0.$$

$$\Leftrightarrow x = 9/2 \qquad (2)$$

So $x = 9/2$ is a stationary point of $D(x)$. This is a minimum stationary point as there is only a single stationary point and it clear there is no maximum value for $D(x)$, consider taking points on the curve with large x-value. (2)

So the point on the curve closest to $(5,0)$ is the point (1)

$$\left(9/2, \frac{3}{\sqrt{2}}\right).$$

Q1 (a) $\{x_n\}_{n=1}^{\infty}$

defined by $x_n = \frac{1}{2n-1}$

Claim $\frac{1}{2n-1} \longrightarrow 0$ as $n \longrightarrow \infty$.

<u>Proof</u>

Def of convergence.

$\forall \varepsilon>0 \;\exists N\in\mathbb{N} \quad n\geq N \Rightarrow |x_n - L|<\varepsilon$

Let $\varepsilon>0$ be given.

Consider the required inequality.

$$\boxed{|x_n - L| < \varepsilon}$$

$$\Leftrightarrow \left|\frac{1}{2n-1} - 0\right| < \varepsilon$$

$$\Leftrightarrow 0 < \frac{1}{2n-1} < \varepsilon$$

$$\Leftrightarrow 2n-1 > \frac{1}{\varepsilon}$$

taking reciprocals, $0<a<b \Leftrightarrow \frac{1}{a} > \frac{1}{b}$.

$$\Leftrightarrow \boxed{n > \frac{1}{2}\left(\frac{1}{\varepsilon}+1\right)}$$

This guides us to choose N as any integer greater than $\frac{1}{2}\left(\frac{1}{\varepsilon}+1\right)$. So now If $n\geq N$

$$\Rightarrow n > \frac{1}{2}\left(\frac{1}{\varepsilon}+1\right)$$

$$\Rightarrow \left| \frac{1}{2n-1} - 0 \right| < \varepsilon,$$ as required.

This proves that $\frac{1}{2n-1} \to 0$ as $n \to \infty$.

b) $\sum_{n=1}^{\infty} x_n$ converges to a sum if and only if the sequence of partial sums converges to a limit.

i.e. $$\sum_{n=1}^{\infty} x_n = \lim_{k \to \infty} \left(\sum_{n=1}^{k} x_n \right).$$

If $x_n \not\to 0$ as $n \to \infty$, then the partial sums will not be "settling down" and converging, as they will ~~ala~~ forever be changing significantly.

d). Consider $\sum_{n=1}^{\infty} a_n$

$$a_n = \frac{3n^2+1}{2n^2}$$

$$= \frac{3 + 1/n^2}{2}$$

$\longrightarrow \frac{3}{2}$, as $n \to \infty$ by algebra of limits theorem and $1/n^2 \to 0$ as $n \to \infty$.

So $a_n \not\to 0$ as $n \to \infty$, and so by General Term Test $\sum_{n=1}^{\infty} a_n$ is not convergent.